Question No 5:-

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Function displayMenu()
    Print "1. Register member"
    Print "2. Display Books"
    Print "3. Borrow a book"
    Print "4. Return a Book"
    Print "5. Display member record"
    Print "6. Exit"
    Print "Enter your choice:"
EndFunction

Function registerMember(name, StudentID, membertype)
    Print "Enter name:"
    Input name
    Print "Enter Student ID:"
    Input StudentID
    membertyp = 0
    while membertype != 1 AND membertype != 2
        Print "Enter membership type (1=Student, 2=Faculty):"
        Input membertype
        if membertype != 1 AND membertype != 2
            Print "Invalid"
EndFunction

Function displayBook() (BookIDs, bookTitles, availableCopies,
                        totalBooks)
    Declare i
    Print "ID    Title"
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for i = 0 to totalBooks-1
    print BookIDs[i], bookTitles[i], availableCopies[i]
EndFunction

Function borrowBook (bookIDs, bookTitles, availableCopies, totalBooks,
                     borrowedBookIDs, borrowedCount, membertype)
Declare id, days, i, pos
    days=0
    pos = -1
    if borrowedCount == 3
        print "You already have 3 books"
        stop
    print "Enter Book ID: "
    Input id
    for i = 0 to totalBooks-1
        if bookIDs[i] == id
            pos = i
    if pos == -1
        print "Invalid Book ID"
        Break,
    if availableCopies[pos] == 0
        print "No copies"
        Break
    while day <= 0
        print "Enter days you want to keep it: "
        Input days
        if days <= 0
            print "Days must be positive"
    if (membertype == 1 AND days > 14) OR
        (membertype == 2 AND days > 30)
    print "a fine applies after the allowed days"
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borrowedBookID[borrowedCount] = id
borrowedCount = borrowedCount + 1
availableCopies[pos] = availableCopies[pos] - 1
print "You borrowed" and bookTitles[pos]
EndFunction
uction returnBook (bookID, availableCopies, totalBooks,
                   borrowedBookIDs, borrowedCount, membertype
                   , totalfine) returns fine
Declare id, days, i, j, pos, stock, allowed, rate,
        fine, code
days = 0
pos = -1
stock = -1
allowed = 30
rate = 10
fine = 10
if membertype == 1
   allowed = 14
   rate = 20
print "Enter BookID to return: "
input id
for i = 0 to borrowedcount-1
   if borrowedBookIDs[i] == id
      pos = i
if pos == -1
   print "you have not borrowed this book"
   return 0
for j = 0 to totalbooks-1
   if bookIDs[j] == id
      stock = j
while days <= 0
   print "enter days the book was kept"
```

```
Input days
if days <=0
    print "Days must be positive"
availableCopies[stock] = availableCopies[stock] + 1
if days > allowed
    fine = (days - allowed) * rate
    totalfine = totalfine + fine
    print "Late return Fine : Rs. fine"
    print "Enter membership code. "
    Input code
    if code == "LIB2026"
        fine = fine/2
        print "50% wavier appleid"
        print "Fine payable : Rs. fine"
    else
        print "Returned on time"
    for i = pos to borrowedCount - 2
        borrowedBookIDs[i] = borrowedBookID[i+1]
    borrowedCount = borrowedCount - 1
    return fine
End Function
Function displayMemberRecord(name, studentID, membertype
                , borrowedBookIDs, borrowedCount,
                totalfine, fineAfterWavier)
Declare i
print "Name: " and name
print "Student ID: " and studentID
if membertype == 1
    print "Membership type: Student"
else
    print "Membership type: Faculty"
print "Borrowed Books:"
for i = 0 to borrowedCount - 1
    print borrowedBookIDs[i]
else
```

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    print " Final fine payable : Rs. fineAfterWavier "
EndFunction
Begin
  BookIDs = 101, 102, 103, 104
  BookTitles = " Programming in C ", Data Structures, " Database
             ", " Matikla "
  Available Copies = 5, 3, 0, 4
  declare name, student ID, member type, borrowedBookIDs
                                          [3]

  student ID = 0
  membertype = 0
   borrowedcount = 0
    total fine = 0
    fine after Wavier = 0
     registered = 0
     choice = 0
  while choice != 6
       displayMenu()
       Input choice
    if choice >= 3 AND choice <= 5 AND registered == 0
         print " Please register first "

    Switch choice
      case 1:-
         if registered == 1
         print " A member is registered "
      else
```

registerMember(name, studentID, membertype)
registered = 1

case 2:

displayBooks(bookIDs, bookTitles, availableCopies, 4)

Case 3:-

borrowbook(bookIDs, booktitles, availablescopies, 4, borrowedBookIDs, borrowedcount, membertype)

Case 4:-

~~fineAfterWa~~

fineAfterwaiver = fineAfterWaiver + returnBook(bookIDs, availableCopies, borrowedBookIDs, borrowedcount, membertype, totalfine)

Case 5:-

displayMemberRecord(name, studentID, membertype, borrowedBookIDs, borrowedcount, totalfine, fineAfterWaiver)

case 6:

print "Goodbye"

default:

print "Invalid choice, try again"



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